When Portfolio Rewards Miss the Target: Exact Mismatch and Policy Alignment

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Abstract

A risk-sensitive portfolio reward need not implement the investor's stated objective. We examine this difference in two settings. For independent returns, we derive the exact leading policy discrepancy that remains after optimally calibrating a quadratic wealth-increment reward: a common scalar penalty cannot recover terminal mean–variance equilibrium when interest is nonzero and the horizon exceeds one period. For constrained allocation with turnover costs, the mean–variance auxiliary identity shows why risk and cost coefficients must be calibrated together. A walk-forward study of ten US industry portfolios over 432 months confirms fitted-policy recovery across 17 settings. Scalar calibration and a log-minus-variance reward retain allocation differences, while an adapted mean–variance policy iteration method matches the reference. Better alignment does not yield a statistically clear realised performance gain. The analysis separates the economic meaning of a reward from optimisation accuracy and investment performance, allowing each to be checked directly.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → *Economics*.

Keywords

portfolio allocation, reward alignment, mean–variance, reinforcement learning, transaction costs

1 Introduction

Suppose an investor asks for mean return minus a variance penalty. Replacing that preference with return minus a squared-return penalty looks harmless, but it also penalises the square of the mean. A learner can optimise the resulting reward accurately and still choose the wrong policy for the stated preference. Interest, trading constraints and transaction costs make the distinction harder to resolve by tuning one coefficient.

Financial learning uses several legitimate objectives. Jiang et al. [10] combine log return and variance; Simon et al. [12] optimise investor utility directly; Cong et al. [4] allow whole-trajectory objectives. Their coexistence makes the reference objective essential: policy disagreement is meaningful only after the intended preference has been specified.

We study the map from reward to policy. First, an independent-return benchmark gives an exact expression for the discrepancy that remains after the best scalar calibration of a wealth-increment reward. Second, a constrained historical study shows how a risk adjustment also changes the effective turnover penalty. Joint calibration restores the fitted preference through the established mean auxiliary used in MVPI [13]. The historical test then asks a separate question: does this alignment improve realised outcomes?

The answer is instructive. Calibration recovers the fitted policy, yet the main realised MV differences are not statistically clear. A useful explanation of a portfolio learner should therefore identify its induced preference as well as its prediction or backtest performance.

2 Related work and comparison design

Objective-aware financial learning. Simon et al. [12] use deep parametric portfolio policies and US equity characteristics. Their Section 4.1, Eq. (17), specifies centred mean–variance utility; Section 3.6, Eqs. (14)–(16), treats turnover costs. Their Table 4 studies alternative preferences. This is a positive example of objective-aware learning, and our fitted covariance-MV reference shares this preference, not their architecture or empirical universe. Gollart and Okhrin [8], Section 4.2.1, Table 1, use zero intermediate rewards and terminal wealth utility, with modified DDPG tested against an analytical Heston–Nandi benchmark. Known-policy validation therefore predates our work. Chaouki et al. [3], Section 3, likewise compare learned and known controls.

Additive rewards and trajectory objectives. Jiang et al. [10], Eq. (8), combine net log return with a variance penalty. Eq. (19) adds soft constraint penalties, and their DPG architecture uses CNN/WaveNet representations. Their Table 2 compares investment horizons across US and Canadian stock universes. We adapt the log-minus-variance preference to the same monthly estimates and hard simplex used by every comparator. We use a first-order turnover charge instead of their log cost factor. The common environment isolates the effect of the reward; the architecture and data differ from those in their study. Jiang et al. [9], Eq. (4), penalise variance and trading costs around a portfolio-*value* term, rather than an excess-return increment. Their TD3/PPO/DDPG comparisons motivate studying risk and cost coefficients together, but are not reproduced here. Cong et al. [4], Eqs. (19)–(20), optimise differentiable trajectory objectives, including a window Sharpe ratio. Such objectives are not presumed equivalent to terminal MV.

Mean–variance control and transformations. Precommitment [11] and equilibrium [1, 6] solve different time-inconsistent control questions. The distinction cannot be settled by a backtest ranking. Czichowsky [5], Lemma 3.2 of the inspected preprint, already gives a discrete continuation-covariance hedge. Björk et al. [2], Propositions 8.1 and 10.2 of the long 2016 preprint, give equivalent control objectives and terminal-unit variance penalisation. Mean–variance policy iteration (MVPI) [13] uses an auxiliary-variable transformation that contains our zero-interest calibration. Our question is more specific: how much policy discrepancy remains after the best scalar calibration of the raw increment reward? The next section gives its exact leading coefficient.

3 An exact test of terminal-objective alignment

3.1 Market, reference and discrepancy

Consider independent excess returns $P_t \in \mathbb{R}^n$ with constant mean $m \neq 0$ and positive definite covariance Σ. Wealth evolves as

$$X_{t+1} = sX_t + P_t^\top u_t, \qquad X_0 = x_0, \tag{1}$$

with unrestricted risky dollar positions u_t and constant gross risk-free return $s > 0$. The terminal objective is $J = \mathbb{E}X_T - \varphi \operatorname{Var} X_T$, $\varphi > 0$. The time-consistent equilibrium policy is

$$u_t^{\mathrm{eq}} = \frac{\Sigma^{-1} m}{2\varphi H_t}, \qquad H_t = s^{T-t-1}. \tag{2}$$

It is not the initial-date precommitment maximiser. Define $M = \Sigma + mm^\top$, $d = M^{-1}m$, $\nu = m^\top d < 1$ and $b = [2\varphi(1-\nu)]^{-1}$. The reward family is $r_t = \Delta X_t - \kappa(\Delta X_t)^2$, with one constant $\kappa > 0$. Let π^κ be its exact additive-reward optimum, and measure

$$D^2 = \sum_{t=0}^{T-1} \mathbb{E}_{\pi^\kappa} \|\pi^\kappa(t, X_t) - u_t^{\mathrm{eq}}\|^2. \tag{3}$$

This is an occupancy-dependent discrepancy, not a symmetric policy metric.

3.2 What scalar calibration cannot repair

Theorem 1. *For fixed $T \geq 2$, as $s \to 1$,*

$$\min_{\kappa > 0} D = c_T |s-1| + O((s-1)^2), \tag{4}$$

$$c_T = \|d\| b \sqrt{\frac{T(T^2-1)}{3} + \frac{\nu(1-\nu)T(T-1)}{2}}. \tag{5}$$

At $s = 1$ exact matching holds for $\kappa = \varphi(1-\nu)$. For $s \neq 1$, $T \geq 2$, exact matching along the surrogate occupancy is impossible.

Proof. Write the Bellman value $W_t(x) = p_t x^2 + q_t x + c_t$, with $p_T = q_T = 0$ and $g_t = \kappa(s-1) - sp_{t+1}$. Completing the square gives $u_t = \alpha_t x + \beta_t$ and

$$\alpha_t = -\frac{dg_t}{\kappa - p_{t+1}}, \qquad \beta_t = \frac{d(1+q_{t+1})}{2(\kappa - p_{t+1})}, \tag{6}$$

$$p_t = \frac{\kappa p_{t+1} - (1-\nu)g_t^2}{\kappa - p_{t+1}}, \quad 1 + q_t = (1+q_{t+1})\left(s - \frac{\nu g_t}{\kappa - p_{t+1}}\right). \tag{7}$$

Induction gives $p_t \leq 0$, so the action Hessian is negative definite and these are global Markov optima. Set $e = s-1$, $h = (2\kappa)^{-1}$. The ratios p_t/κ and q_t are independent of κ, hence pathwise $X_t = L_t(e)x_0 + hZ_t(e)$. Consequently $D^2 = F(e,h)$ is exactly quadratic in h, with analytic coefficients near zero and $F(0,h) = T\|d\|^2(h-b)^2$. Its global minimiser is analytic and remains positive near zero.

Put $\kappa = \varphi(1-\nu)(1+we)$. The leading action difference is de times

$$-X_t + b[(T-t-1)(2-\nu) - w].$$

At $e = 0$, $\mathbb{E}X_t = x_0 + tvb$ and $\operatorname{Var} X_t = tb^2\nu(1-\nu)$. Minimisation yields $w^* = (T-1)(1-\nu) - x_0/b$. The centred means are $b(T-1-2t)$; summing their squares and variances gives the displayed c_T^2. Analyticity then gives the stated remainder after taking square roots.

Finally, matching at the first date forces a nonzero equilibrium action and positive subsequent wealth variance. At the final date the surrogate has slope $-d(s-1)$ whereas equilibrium is constant. Almost-sure equality is impossible when $s \neq 1$. □

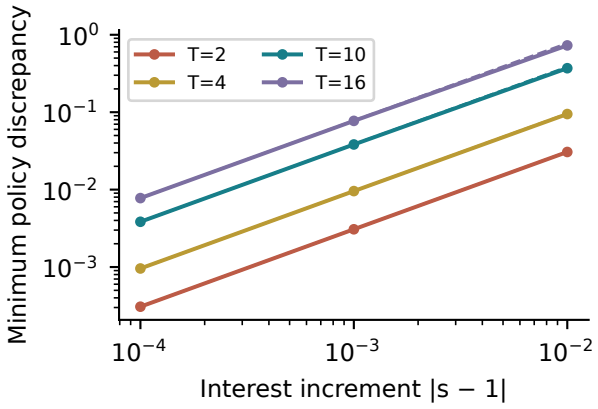


Figure 1: Exact minimum discrepancy (points and solid lines) and local prediction $c_T|s-1|$ (dashed lines). Here $x_0 = \varphi = 1$, $m = (.06, .04, .02)$, volatilities $(.18, .12, .08)$, and correlations $(\rho_{12}, \rho_{13}, \rho_{23}) = (.30, -.10, .05)$. This concerns terminal equilibrium, not the fitted historical reference.

The coefficient grows as $T^{3/2}$ for large T, but the expansion is for fixed T; no uniform joint interest–horizon limit is claimed. The obstruction concerns this reward family, not all additive objectives.

3.3 A reference-preserving reward

Allow moments to vary by date while returns remain independent. Define $Y_t = H_t P_t^\top u_t$ and use

$$r_t^{\mathrm{design}} = Y_t - \varphi(1-\nu_t)Y_t^2. \tag{8}$$

The expected reward is strictly concave. Its maximiser is

$$u_t = \frac{\Sigma_t^{-1} m_t}{2\varphi H_t}.$$

Future rewards admit state-independent maximisers, so backward induction recovers the equilibrium. This construction specialises the equivalent-objective approach of Björk et al. [2] and changes the optimum of the original raw reward. Dependent returns require continuation terms, so Eq. (8) is not applied as a historical multiperiod theorem.

4 Constrained alignment with turnover costs

Historical implementation adds constraints and costs. On a fixed past-only estimated state, let C be the simplex, p the pre-trade allocation and $c(u) = \|u - p\|_1$. Define the fitted one-period preference

$$F(u) = m^\top u - \varphi u^\top \Sigma u - \lambda c(u), \quad u \in C. \tag{9}$$

Its unique optimum u^* is a *fitted reference*, not the unknown true market optimum. This section does not extend Theorem 1 to estimated or dependent markets.

Proposition 1. *Let $y^* = m^\top u^*$ and $A = 1 + 2\varphi y^* > 0$. The second-moment reward objective*

$$Q(u) = m^\top u - \frac{\varphi}{A} u^\top M u - \frac{\lambda}{A} c(u) \tag{10}$$

has the same unique maximiser as F.

Proof. For all feasible u,

$$AQ(u) = F(u) - \varphi(m^\top u - y^*)^2 + \varphi(y^*)^2. \tag{11}$$

At u^* the square vanishes and F is maximal. Since $A > 0$, the assertion follows. □

The argument applies to any feasible set with a global maximiser. Convexity and positive definite covariance ensure uniqueness here. The identity follows the mean auxiliary used in MVPI [13].

If only the risk coefficient is divided by A, while the cost coefficient stays λ, the right-hand side of Eq. (11) gains the extra term

$$-2\varphi y^* \lambda c(u). \tag{12}$$

When $y^* > 0$, this increases the effective cost preference. It need not imply monotone realised turnover on the strategy's evolving path. The unconstrained formula $\varphi(1 - \nu)$ likewise need not calibrate a constrained optimum.

Joint calibration uses u^* and therefore serves as a certificate of alignment with a known reference. As a constructive prior-art comparator, we also alternate maximising

$$(1 + 2\varphi y) m^\top u - \varphi u^\top M u - \lambda c(u) \tag{13}$$

and setting $y = m^\top u$. This is an adaptation of the MVPI mean auxiliary to the constrained one-period problem. Including the term $-\varphi y^2$ gives a jointly strictly concave objective, since its quadratic block has Schur complement $\Sigma \succ 0$. Exact block updates converge to the same target. This provides a constructive comparator with the same optimum.

5 Historical study

5.1 Data and protocol

We use Kenneth French's ten value-weighted US industry portfolio monthly total returns and the Fama–French risk-free series [7]. The archived downloads have a recorded SHA-256 manifest. Estimation data start January 1970; evaluation covers January 1990–December 2025 (432 months). This is a retrospective study on the August 2026 revised vintage, not a point-in-time constituent dataset. Industry portfolios are research portfolios, not directly tradeable funds; their internal reconstruction costs are not included. The provider's CRSP processing and risk-free source changes also form part of this data vintage.

For month t, use only the previous 120 months of excess returns to estimate m and the unbiased covariance, shrunk 10% toward its diagonal. All strategies are long-only and fully invested. The primary settings are $\varphi = 5$ and $\lambda = 0.001$ (10 basis points per unit of L1 turnover). Pre-trade weights drift with realised gross asset returns. Net portfolio return uses the explicit first-order convention $u_t^\top R_t - \lambda \|u_t - p_t\|_1$, not a market-impact model or exact intramonth execution equation. The experimental protocol fixed these choices before strategy outcomes were inspected.

5.2 Comparators and measurements

We solve the covariance-MV target, raw second moment ($\kappa = \varphi$), unconstrained scalar calibration ($\kappa = \varphi(1 - \nu)$), reference-based risk-only calibration, joint calibration, and MVPI adaptation. Equal weight and minimum variance provide conventional portfolio controls. Log utility and log-minus-variance adapt the reward preference in Jiang et al. [10]; every method uses the same monthly data and feasible set. The comparison tests induced preferences under common estimates, not which published learning architecture is best.

For alignment, all methods face the target strategy's *same pre-trade state*. We report Euclidean allocation distance and fitted regret $F(u^*) - F(u)$. Each strategy also follows its own pre-trade states for realised performance. This avoids conflating state divergence with the local reward comparison. Quadratic solutions have independent linear-programme first-order certificates; the largest saved common-state certificate is 4.27×10^{-9} in objective units. The observed agreement of calibrated policies is considerably tighter than the acceptance tolerance.

Realised metrics are monthly excess-return MV, annualised excess Sharpe, return, volatility, turnover and drawdown. Uncertainty for paired MV differences uses 1,000 circular moving-block bootstrap resamples, blocks of 12 months, seed 8128. These intervals describe the observed strategy-return sequences, not full estimator-refitting or model-selection uncertainty. Robustness spans 17 pre-specified combinations: $\varphi \in \{2, 5, 10\}$ and cost $\{0, 10, 25\}$ bp at the primary estimator, plus windows $\{60, 120, 240\}$ and covariance shrinkage $\{0, 0.1, 0.5\}$ at the primary preferences. The overlapping primary case is counted once.

5.3 Results

Table 1 shows that raw second-moment optimisation has mean allocation distance 0.00843. Unconstrained scalar calibration reduces this to 0.00540; reference-based risk-only calibration retains 0.00510. Joint calibration and MVPI agree with the fitted reference to numerical precision. The adapted log-minus-variance objective has distance 0.01286; log utility alone is much farther away. The differences measure departure from the specified MV preference; log utility represents a different investment objective.

Across all 17 settings, the largest mean joint-calibration distance is 1.54×10^{-7}; raw mean distances range from 0.00157 to 0.03682. Thus constraints and trading costs do not break the joint identity, while unconstrained scalar calibration is not a universal repair. Figure 2 relates alignment to historical outcomes, and Figure 3 gives robustness and uncertainty.

Joint calibration has realised monthly MV lower than raw by 1.315 bp, with 95% interval $[-3.318, 0.343]$ bp. Against scalar calibration the difference is 0.385 bp. Its interval is $[-0.570, 1.351]$ bp. Against risk-only calibration the difference is -0.131 bp, with interval $[-0.889, 0.634]$ bp. All include zero. Minimum variance has both a better realised MV score and higher Sharpe than the fitted target in the primary sample. This illustrates estimation risk: precise implementation of estimated preferences cannot guarantee the best realised allocation. Equal weight has higher Sharpe than the target but lower realised MV, so Sharpe ranking alone does not answer even the ex-post MV comparison.

Table 1: Primary historical comparison, 1990–2025. Distance and fitted regret use common states; realised metrics use each strategy's own path. Gap and MV score are monthly basis-point units. Calibration improves model alignment; the table does not establish superior investment performance.

Method	Mean allocation distance	Fitted MV gap (bp)	Excess Sharpe	Realised MV (bp)
Target	0.00000	0.0000	0.570	-8.497
Raw second moment	0.00843	0.0148	0.573	-7.182
Unconstrained scalar	0.00540	0.0043	0.568	-8.882
Risk-only calibration	0.00510	0.0033	0.571	-8.365
Joint calibration	0.00000	0.0000	0.570	-8.497
MVPI adaptation	0.00000	0.0000	0.570	-8.497
Log minus variance	0.01286	0.0287	0.572	-7.123
Log utility	0.54848	39.7855	0.344	-133.295
Equal weight	0.43495	46.2626	0.622	-11.893
Minimum variance	0.23843	9.0252	0.645	5.274

5.4 Separating optimisation from reward design

A finite-budget experiment assigns independent softmax policy logits to each of the 432 historical common states. Adam uses learning rate 0.05, initial logits $N(0, 0.1^2)$, five matched seeds and 50, 250, 1,000 and 5,000 gradient steps. Each objective uses only its state's lagged estimates. Independent logits isolate optimisation error on each fitted state. They do not test generalisation across states, as a predictive neural policy would.

Let u_r maximise the specified reward and $\widehat{u}_r$ be the finite-budget allocation. We report the nonnegative own-reward optimisation gap separately from

$$F(u^*) - F(\widehat{u}_r) = [F(u^*) - F(u_r)] + [F(u_r) - F(\widehat{u}_r)]. \tag{14}$$

The second bracket is a *signed* target-value difference; it is not the own-reward optimisation gap. At 5,000 steps, mean own-reward optimisation gaps remain approximately 4×10^{-6}, exceeding the exact raw design gap of 1.48×10^{-6}. Thus finite-budget error can dominate the reward discrepancy. Figure 4 shows why these must be kept separate: improving the raw-reward optimiser does not remove its exact design gap. The target and joint reward have the same limiting optimum but can exhibit different finite-budget trajectories because their objective geometries differ.

6 Implications and scope

The analytical result is proved for independent returns and unrestricted dollar positions. The historical identity is proved for a fitted one-period problem with a positive calibration factor; its numerical application uses constrained allocations and lagged data. Neither result establishes a true historical terminal-MV equilibrium under dependence. The historical returns themselves need not be iid, but the rolling one-period reference does not identify their complete conditional law or continuation hedge.

The practical procedure is to state preferences before training, identify an appropriate reference, solve or approximate the reward optimum, and audit both on identical states. If a second-moment surrogate is used with turnover costs, Eq. (12) identifies a concrete source of preference drift. A reference-based joint calibration or existing auxiliary-variable method removes that fitted discrepancy. Subsequent learning and historical performance remain separate empirical questions.

Scalar tuning has a precise limit in the independent terminal-wealth problem. With constraints and turnover costs, joint calibration restores the fitted one-period preference, while the realised outcome still depends on estimation quality. These results make reward semantics testable before interpreting a learned policy or comparing backtests. Extending the audit to dependent terminal equilibria and point-in-time investable portfolios would require a richer reference model.

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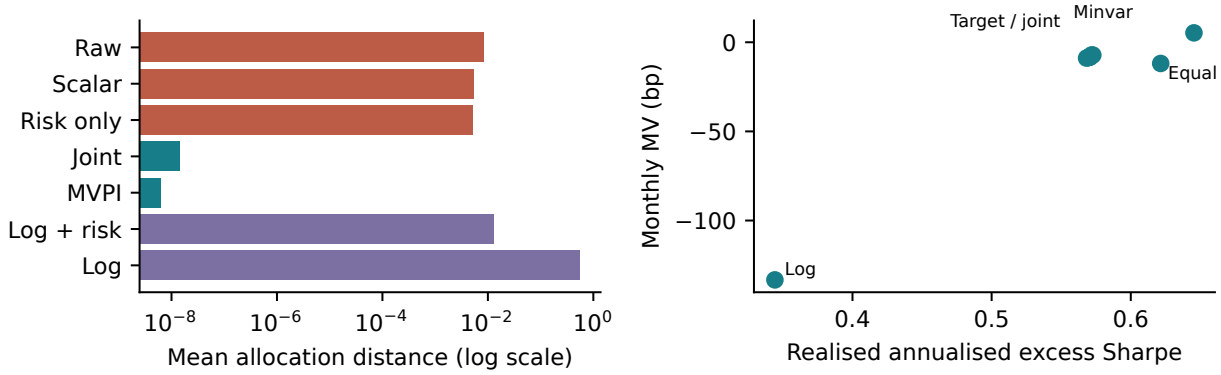


Figure 2: Left: mean common-state allocation distance; the logarithmic axis exposes numerical agreement for joint calibration and MVPI. Right: realised Sharpe and monthly MV need not rank policies alike. Equal weight has higher Sharpe but lower MV than the fitted target.

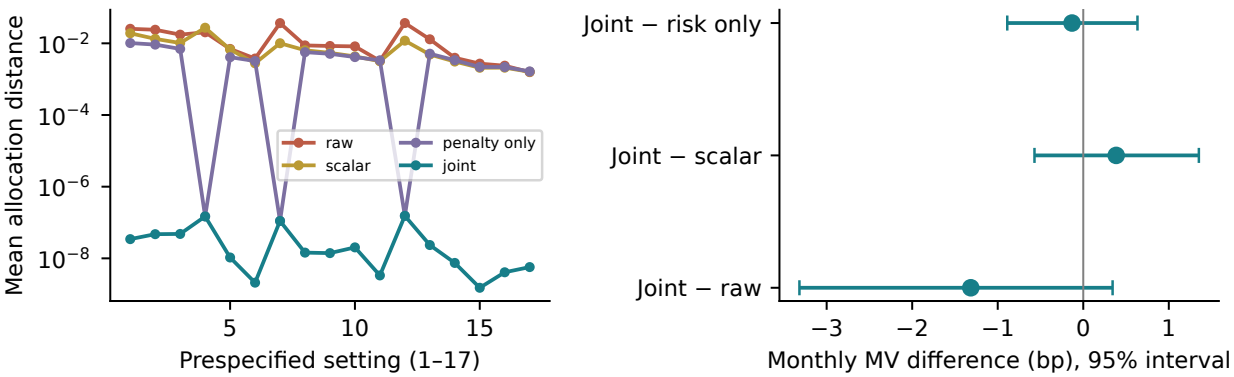


Figure 3: Left: alignment across the 17 prespecified settings, ordered by window, risk penalty, cost and shrinkage. Right: paired 12-month-block bootstrap intervals all include zero. Exact fitted alignment does not imply a statistically clear realised MV advantage.

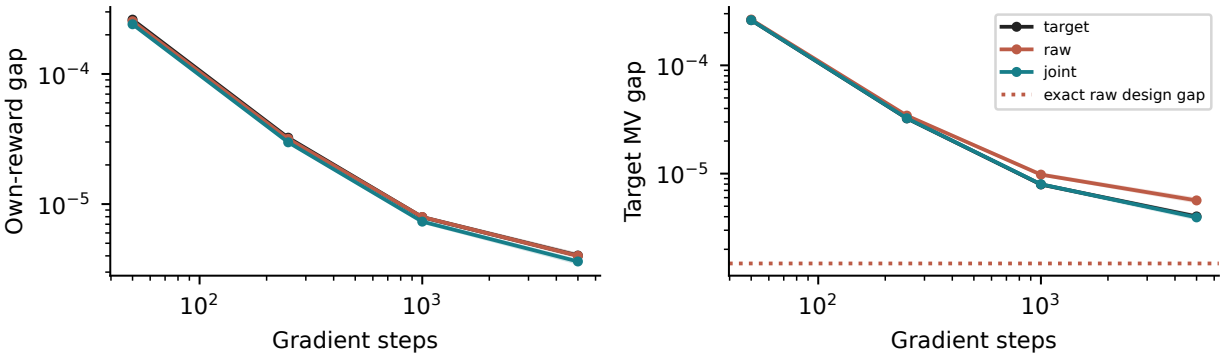


Figure 4: Finite-budget softmax optimisation on historical fitted states. Lines are five-seed means, shading the seed range. Left uses each method's own reward; right uses the common MV target. The dotted line is the exact raw reward-design gap. This diagnostic measures optimisation accuracy, not generalisation.

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